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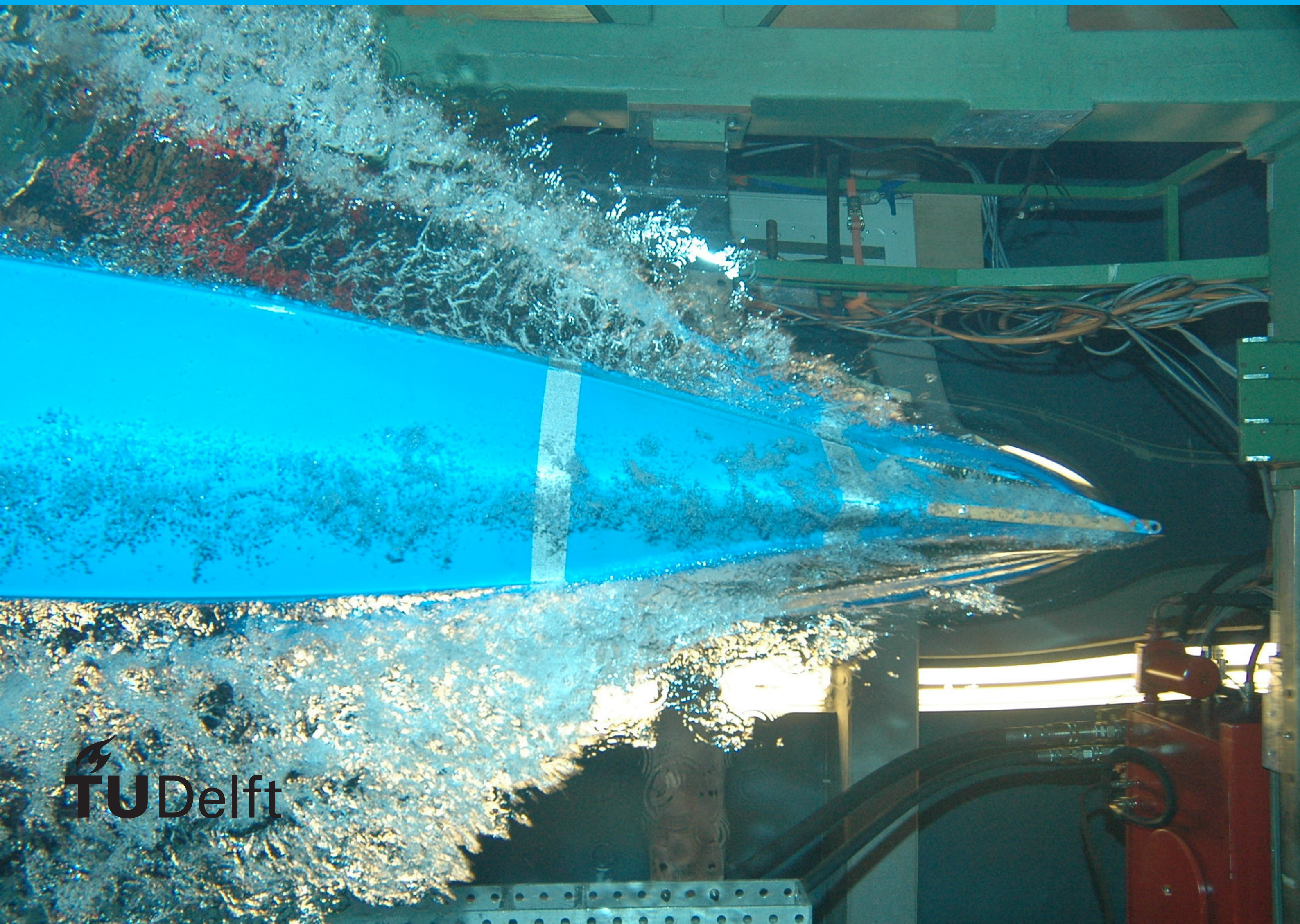

TU Delft

Title
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TU Delft

Preface

Preface...

J. Random Author
Delft, January 2013

Contents

1 Introduction

Sedimentary deposits form a precious commodity in our current society due to the various resources which can be extracted from them. These resources include minerals, ground water, geothermal heat and hydrocarbons (e.g. Morris and Ramanaidou, 2007; Donselaar et al., 2017; Donselaar et al., 2015; Brekke et al., 2001). Moreover, in recent years studies have been looking at storage possibilities of carbon dioxide and hydrogen in this subsurface environment (Zhou et al., 2020; Heinemann et al., 2021). Besides the increase in subsurface applications we have to consider an increasing world-wide population combined with an ever growing demand for minerals needed for the energy transition, as well as the energy itself. As a result the load on fluvial sedimentary deposits will only increase while the distribution of the commodities themselves only gets sparser and of lower quality (e.g. Spittler et al., 2020; Calvo et al., 2016). This calls for a better understanding of the subsurface environments the resources are located in, unlocking the possibility to better predict resource management. In the exploration process, after collecting data and defining a geological conceptual model, 3D geomodels are generated which are significant to predicting reservoir (heat)flow behavior and resource spread (SONG et al., 2022; Royer et al., 2015).

1.1. Problem definition

Geological 3D modelling represents one's hypotheses about the most likely rock mass configuration, constructed from limited and sparse data Koltermann and Gorelick, 1996. This low-data environment makes it necessary to construct 3D geomodels based on limited data through interpretations of a conceptual model. After the definition of a conceptual model, several modelling approaches may be applied, leading to several limitations based on the approach taken:

- **Limiting input of geological knowledge:** When developing a geological realization, the model choice is important to consider. often the models rely on a weak-geological prior as their input is restricted to distributions of components (Rongier and Peeters, 2025). Though methods such as object modelling are able to recreate complex patterns, more complex relationships cannot always be captured (Sun et al., 2023). Moreover, there is a need for applying a multi-conceptual model approach into the workflow to minimize single-conceptual model uncertainties (Enemark et al., 2019). Developing a full scale model for each conceptual will take time as geological cases with different architectures and dimensions have to be defined, making this approach labor-intensive.

- **Uncertainty propagation:** The workflow of Geological modelling is divided into multiple steps with uncertainty added as each step is undertaken, either through data inputs or generated through steps (Høyer et al., 2024). The type of uncertainties have thus far been categorized into subjective and objective uncertainties (Fookes, 1997): the objective uncertainties may relate to uncertainties in data while subjective uncertainties may arise through interpretations. Due to beliefs and biases based in the interpretations, the uncertainties arising from them may propagate to the final 3D model. Therefor modelling requires professional judgment, balancing objectivity and subjective intuition (Zorzi et al., 2025).

As Caers, 2011 noted that theoretically, the uncertainty that arises from subjective sources may be infinite in magnitude. As a result, the question arises on how the current workflow can be optimized such that the uncertainties may be reduced.

1.2. Research Gap and Opportunity

The application of machine learning techniques incorporated in geological modelling is not a completely new approach, as shown by Demyanov et al., 2008. However, recent developments in the computer science field and computing power have unlocked a new type of machine learning algorithms:

Deep Generative Models. Recent work in the field op geological modelling has demonstrated the potential for several deep generative modelling techniques to produce geological correct realizations of the subsurface. Variational Auto-Encoders, Generative Adversial Networks and 3D Latent Diffusion Models (e.g., Laloy et al., 2017; Bhavsar et al., 2024; Song et al., 2023; Di Federico and Durlofsky, 2025). The advantages of these models stems from the fact that, when trained, 1) the models are able to generate realizations much faster than traditional process-based models and 2) given enough training data are able to capture non-stationarity within the subsurface (Rongier and Peeters, 2025). With regards to the possible improvements with could be made for integration of machine learning in the geological modelling field (Rongier and Peeters, 2025), stated the following: “We now need to move towards more detailed studies on larger case studies closer to field applications to further assess how the whole workflow performs.” With Di Federico and Durlofsky, 2025 made a similar comment: “Finally, the application (and extension as required) of the overall methodology to real cases should be performed.” With this in mind the goal of this project should be clear: assess the performance of deep generative modelling in real world cases.

1.3. Objective and Research Questions

the main goal of this thesis is to **assess the practical feasibility of using a Generative Adversarial Network (GAN) to generate realizations of a real-world reservoir by conditioning it to well data**. The outcome of this thesis could indicate whether GAN’s can be used to model the subsurface in various settings and if other Deep Generative modelling techniques can be considered. Hence, he main research question of this work is:
To what extend can a Deep Generative Modelling (DGM) be used to create geologically plausible realizations, of the Delft Sandstone reservoir while conditioned to sparse real world data?
In order to help anwering question, various subquestions are formulated:

1. How should the Delft Sandstone’s geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modeling?
2. Which DGM methods is best suited for 3D geological modelling.
3. How well does the conditioned model produce geologically realistic structures and is it able to produce new, previously not seen structures?

   a. Which items is the model able to reproduce when generating samples? Are structures lost in the process?
4. Which evaluation metrics can be used and how does the final model compare to realizations generated by traditional modelling approaches?

1.4. Thesis outline

The remainder of this thesis structured as follows:

- **Chapter 2:**
- **Chapter 3:**
- **Chapter 4:**
- **Chapter 5:**
- **Chapter 6:**

2 Background

The data used in this report comes from the Delft Geothermal Reservoir, located at the Delft University of Technology Campus in the Netherlands. As of Februari 2026 there is a doublet actively producing geothermal heat from the reservoir (Beerda, 2026). The doublet consists of a producer (**DEL-GT-01**) and injector(**DEL-GT-02-S2**), where the producer is drilled up to a TvD of ± 2337 m and the injector to a TvD of ± 2100 m (Vardon et al., 2024). Designed to be a research as well as a commercial project, in both wells a large number of logs and cores were collected with additional installment of monitoring instruments both in the wells as on the surface (Vardon et al., 2020 and Vardon et al., 2024). The targeted layer of this project is the Delft Sandstone, a member of the Lower Cretaceous Nieuwerkerk formation in the West Netherlands Basin (WNB) (Donselaar et al., 2015).

2.1. Geology of the Delft Reservoir

The WNB is a 60 km wide transtensional basin in the SW of the Netherlands with NW-SE trending structural highs and depressions (Ziegler, 1990 as cited in Donselaar et al., 2015). The WNB entered an active rifting stage around the Middle-Jurassic (Bodenhausen and Ott, 1981 & Den Hartog Jager, 1996 as cited in Willems et al., 2020) leading to the formation of several half-grabens, later filled with terrestrial sediments sourced from neighboring regions (Den Hartog Jager, 1996). the overall trend during the deposition of sediments fills in the halfgrabens led to a change in depositional environment: a transgressing shoreline moving the depositional environment from a fluvial to marine system. The Nieuwerkerk formation consists of three members: the Alblasserdam member, Delft Sandstone member (DSST) and Rodenrijs Claystone member. According to DeVault and Jeremiah, 2002, the Alblasserdam member was deposited in a fluvial environment with stacked fluvial channels and floodplain deposits. The DSST directly overlays the Alblasserdam and consists of thick stacked channel sandstone deposits with high net-to-gross values (Willems et al., 2020; Mijnlieff, 2020 and Bruining, 2023). The DSST is overlain by the Rodenrijs Claystone member which is rich in organics with characteristics of a lagoonal deposit (Willems et al., 2020 & DeVault and Jeremiah, 2002 as cited in Bruining, 2023).
Based on recent research by Bruining, 2023, who analysed the cored section of **DEL-GT-01** and sidewall cores of **DEL-GT-02-S2**, six distinct lithofacies types could be determined. Eventually three facies associations were constructed:

- **FA1:** Fluvial and distributary channel deposits.
- **FA2:** Crevasse spays and levee deposits.
- **FA3:** Overbank deposits and paleosols.

The FA1 facies is characterized by sediments that are primarily light grey fine to medium-grained sandstones that feature low-angle cross-bedding and ripple lamination with unit thicknesses ranging from 5 - 80m. FA2 is composed of well to moderately sorted, very fine to medium-grained sandstones interbedded with very dark grey siltstones and unit thicknesses ranging from 1 to 5 metres. Finally, the FA3 association consists of dark grey to very dark grey siltstone, claystone, and very fine-grained sand deposited in low-energy environments. This association is uniquely characterised by an abundance of organic material, slickensides and millimetre-sized siderite nodules often found at the base of sequences.
The depositional environment in which the DSST member was deposit has been varying in scientific literature. The DSST is attributed to both a fluvial-meandering environment (Donselaar et al., 2015; Vondrak et al., 2018 & Willems et al., 2020) as well as a fluvio-deltaic system (van Adrichem Boogaert and Kouwe, 1993; Weert et al., 2024 & Bruining, 2023). With the report by Bruining, 2023 having anlysed all the available cores of the **DEL-GT-01** and **DEL-GT-02-S2** boreholes leads to the most recent hypothesis that the *cored section* of the DSST geothermal reservoir in Delft originates from a Deltaic

environment of deposition. Only the first 80m of the DEL-GT-01 well is cored, amounting to ±1/4th of the full well, with result that the rest of the section might leap into a different depositional environment.

2.2. Depositional Environment

Deltaic environments of deposition form at the boundary of sea and land. Clastic sediments are supplied by river systems and deposited in a radial form at the shoreline. Several variations of delta's exist based on the dominant process influencing the delta shape: River dominated, Wave dominated or tide dominated delta's (Kolb and Van Lopik, 1966). In order to describe characteristics of the subaerial deltaic deposits, three different subdomain are categorized (e.d. Kolb and Van Lopik, 1966 & Heldreich et al., 2017): Upper Delta plain, Lower Delta Plain and subaquoes Delta...

- **Upper Delta Plain:** The upper delta plain lies above the level of effective saltwater intrusion and is unaffected by marine processes. Most of the sediments comprising this part of the delta plain originate from the migratory tendency of distributary channels, overbank flooding during annual highwater periods, and periodic breaks in the river banks, in which "crevassing" into adjacent lake basins occurs. The major environments of deposition include braided channels, meandering channels (point bars and meander-belt deposits), lacustrine delta fill, backswamps, and flood plains (swamps, marshes, and freshwater lakes).
- **Lower Delta Plain:** The lower delta plain lies within the realm of river-marine interaction and extends landward from the shoreline to the limit of tidal influence. Most commonly in this environment, channels become more numerous and often show a bifurcating or anastomosing type of plan view pattern. Environments between distributary channels comprise the largest percentage of the lower delta plain and consist of actively migrating tidal channels, overbank splays (natural levees), interdistributary bays, bay fills (crevasse splays), marshes, and swamps. The major environmental sequences described consist of bay-fill deposits (interdistributary bay, crevasse splay-natural levee, and marsh) and abandoned distributary-fill deposits.

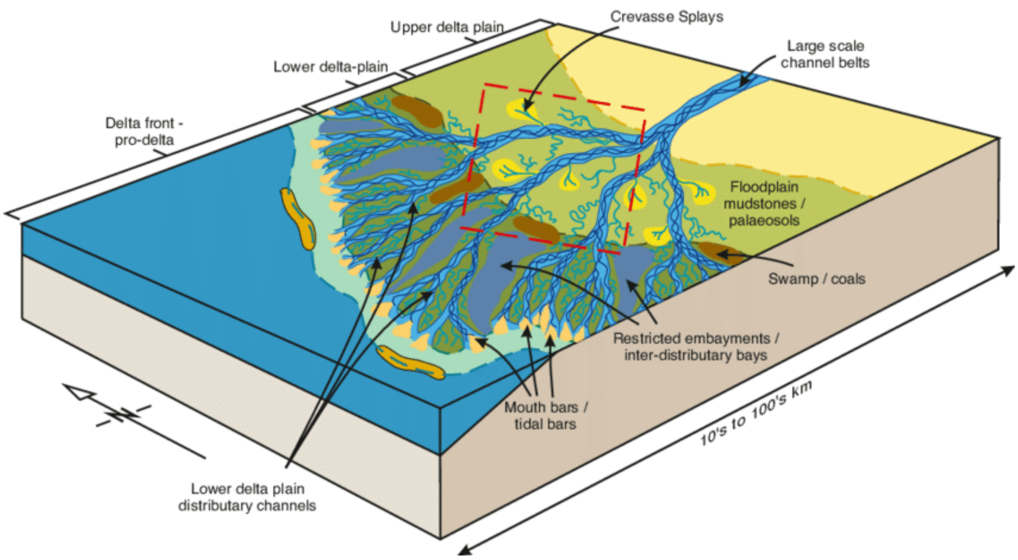

Figure 2.1: Example of depositional model of the Mungaroo Formation (North West Shelf, WA, Australia) highlighting the different geometries and definitions (Heldreich et al., 2017).

2.3. Geo-modelling

The general purpose of 3D reservoir models is to provide a 3D representation of the subsurface which also allows for reservoir simulations to be performed making informed decisions possible. The model can be be updated as new information becomes available, making it essential for predicting reservoir behavior over the course of the reservoir's lifetime (Emerick, 2025 & Ringrose and Bentley, 2016).

Koltermann and Gorelick, 1996 categorized existing models into three different main techniques: object-based, process-based and descriptive/geostatistical models. Where object-based models mostly ignore physics to directly place geological features, discrete models use spatial statistics for interpolation. In contrast process-based models are based on fundamental laws of conservation of mass and momentum to simulate the creation of geological features. Each model has unique advantages and disadvantages. As a result, over the past decades research has been oriented at combining different models to leverage each models advantages creating so called 'Hybrid models' (e.g. Koltermann and Gorelick, 1996; Lopez et al., 2009; Michael et al., 2010).

2.4. Deep Generative modelling

2.5. Summary

3 Methodology

This chapter aim to clarify the complete workflow of this project. This includes the generation on 3D datasets used for training, validating and testing the model which is covered in section 3.1. Section 3.2 described the exact GAN framework used in the project. This includes the facies representation approach as well as the validation function and loss function which are used to monitor the model during training. Finally, section 3.4 introduces the evaluation metrics used to test geological realism in the final realizations.

3.1. Training data generation

In order to train the Deep learning algorythm a large number of training samples representing the ground truth ar needed. Since the subsurface is a 3D domain with a high degree of uncertainty the current method to obtaining a large number of training samples is by simply generating them using a quick algorithm (Rongier and Peeters, 2025; Sun et al., 2023). The approach taken in this project is to start of with three different geological conceptual models, two of which are based on the upper and lower plain delta environment and the final conceptual model has a general deltaic representation. The

3.2. GAN framework

3.3. Conditioning to well data

3.4. Evaluation Framework

4 Results

5 Discussion & Limitations

6 Conclusion & Recommendations

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